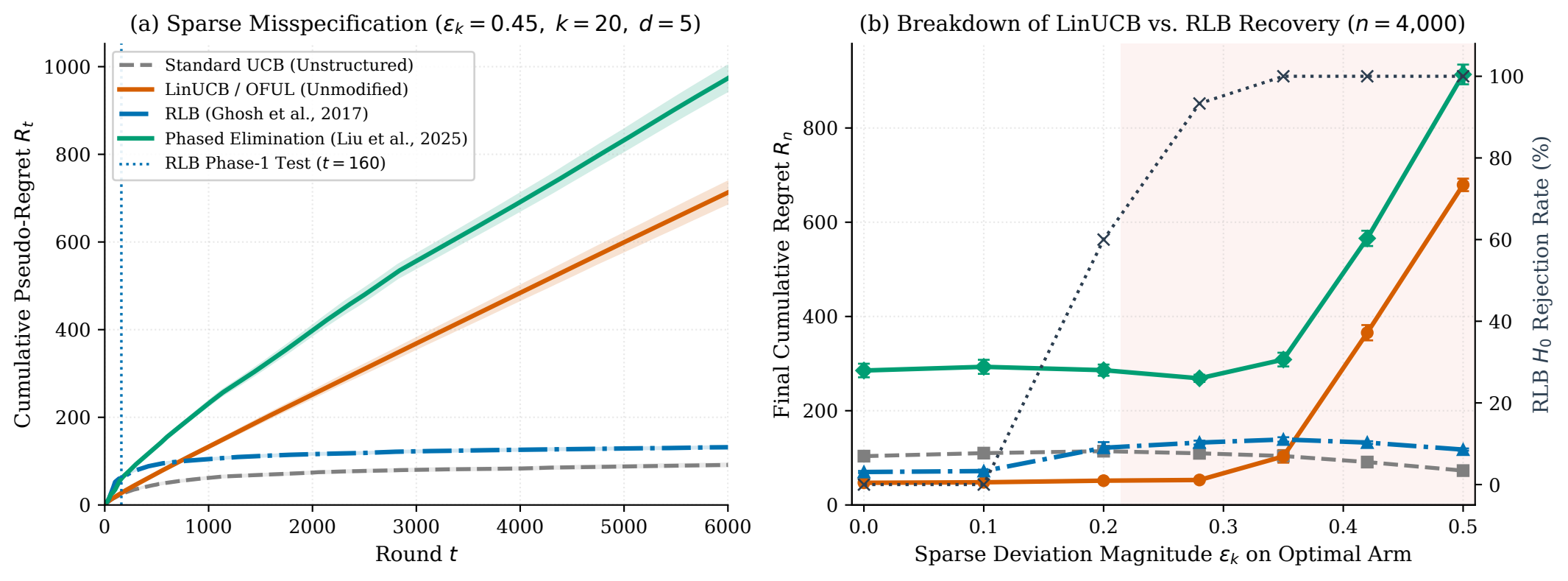

(a) Sparse Misspecification ($\varepsilon_k = 0.45,\ k = 20,\ d = 5$)
Standard UCB (Unstructured)
LinUCB / OFUL (Unmodified)
RLB (Ghosh et al., 2017)
Phased Elimination (Liu et al., 2025)
RLB Phase-1 Test ($t = 160$)
Cumulative Pseudo-Regret R_t
Round t
0
200
400
600
800
1000
0
1000
2000
3000
4000
5000
6000
(b) Breakdown of LinUCB vs. RLB Recovery ($n = 4{,}000$)
Final Cumulative Regret R_n
RLB H_0 Rejection Rate (%)
Sparse Deviation Magnitude ε_k on Optimal Arm
0
200
400
600
800
0
20
40
60
80
100
0.0
0.1
0.2
0.3
0.4
0.5